A model of R&D leadership and team communication: the relationship with project performance

Giles Hirst[1] and Leon Mann[2]

[1]Work and Organisational Psychology Group, Aston Business School, Aston University, Birmingham B47ET, UK. g.hirst@aston.ac.uk
[2]Melbourne Business School, University of Melbourne. 200 Leicester Street, Carlton, Victoria. 3053, Australia. l.mann@mbs.unimelb.edu.au

Industrial research and development (R&D) involves the processing and transformation of new knowledge into a commercially valuable outcome. Communication is an effective mechanism to translate, share and integrate new information into commercial products or processes. We developed a five-factor model of team communication comprising: leadership role performance, team boundary spanning, communication safety, team reflexivity and task communication and tested the model using a one-year longitudinal study. Analyses were conducted on team level data from 56 teams, comprising 350 employees. Independent measures of project performance were obtained from surveys of research managers as well as project customers. Three findings emerged. Different factors predicted different stakeholders' ratings of project performance. Communication safety was the strongest predictor of customer ratings of performance. Boundary spanning is most effective when performed by the project leader not the team.

1. Introduction

The awareness that successful teamwork is dependent on interaction and knowledge sharing amongst team members has led to a proliferation of team communication studies (Marks *et al.*, 2001). Team communication has been found to predict innovation (Ancona and Caldwell, 1992), project performance (Brodbeck, 2001; Keller, 2001) as well as patents and commercialized products (Allen, 1984; Visart, 1976). However despite making considerable progress several questions remain unanswered. Few studies have examined the range of communication behaviours that teams engage in. Even empirical studies that distinguish and measure both external and internal communication are rare (Kivimaki *et al.*, 2000). As a consequence we possess limited insight into the communication behaviours that teams should engage in, nor understand the interaction between different aspects of communication. Further researchers who have studied project teams have done so either from a leadership or team's perspective, with little attention being paid to the interaction between the two. As a consequence divergent prescriptions emerge from different perspectives. For example, leadership studies (cf. Waldman and Atwater, 1994), emphasize that the project leader's authority and seniority is essential to lobby and overcome resistance and interface with the wider organization. Whereas the team and communication network literature (cf. Allen, 1984; Boutty, 2000) articulates that extensive and diverse networks ensure teams are able to acquire scarce resources and develop a comprehensive understanding of sometimes divergent and complex information. Studies which measure both leadership and team communication would reconcile these perspectives.

We seek to improve our understanding of team communication by developing and testing a model of team communication. In particular, we make three contributions to the literature. Firstly, we identify key leader and team communication behaviours. Consistent with previous research, which has identified the importance of role based models (cf. Kim *et al*., 1999; Hooijberg and Choi, 2000) we examine four leadership roles. The four roles were identified by Barry (1991) and refined by Yukl (2002) based upon qualitative observations of engineering and new product development teams. They include leadership boundary spanning, facilitative leadership, leadership which stimulates reflection and innovation as well as task leadership. Corresponding to the leadership roles we identify four team communication processes. These processes include team boundary spanning, communication safety, team reflection and task communication. We summarize and integrate these approaches to develop a five-factor model of team communication (see Figure 1).

Secondly, we seek to address a surprising gap within the literature, that despite the centrality of leaders to team performance, and considerable research examining team and leadership processes, we know surprisingly little about how leaders create and handle effective teams (Zaccaro *et al*., 2001; Pirola-Merlo, Hartel, Mann & Hirst, 2002). Based on recent team leadership research of Zaccaro *et al*. (2001) we test a functional leadership framework. This approach diverges from traditional leadership perspectives (e.g. Fiedler, 1964; Kerr and Jermier, 1978) treating team processes not as moderators but as mediators (Zaccaro *et al*., 2001). The premise behind this framework is that leaders are responsible for managing teamwork and their role is to ensure task accomplishment and group maintenance is adequately managed. Thus leadership behaviours facilitate team communication, which in turn drives effective performance.

Thirdly, we incorporate methodological directions highlighted by Cohen and Bailey (1997) who affirm the need for longitudinal studies that use repeated measures of performance and draw upon multiple ratings of performance. Their conclusions parallel the findings of recent research, such as Hooijberg and Choi's (2000) study of team leadership and Hoegl and Gemuenden's (2001) study of software development teams that have observed stakeholders hold divergent ratings of performance. Thus we survey teams, customers and research managers' ratings of performance. Whilst this contribution is methodological, it reflects an increasing awareness that within organizations there are different constituents of performance and as a consequence different stakeholders may hold different sometimes clashing agenda (West *et al*., 1997). Indeed, Shankman (1999) argues that the identification of different stakeholders' agenda and understanding

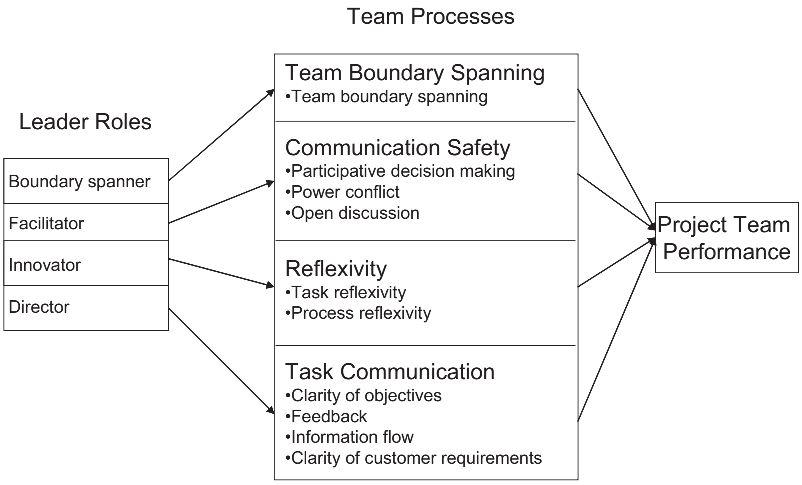

Figure 1. Representation of the leadership and team communication model.

 © Blackwell Publishing Ltd 2004

their performance implications is one of most instrumental challenges facing modern organizations. Thus we implicitly acknowledge in complex decision-making settings, such as the R&D context, there may not be a single performance standard.

2. Model of team communication

2.1. Leadership role performance

Team leadership studies have demonstrated the importance of role based leadership models to R&D performance (Barry, 1991; McCall, 1988; Kim *et al*., 1999). The basic premise of these studies is that team leaders must be competent at performing a diverse array of leadership behaviours. These models tend to measure the extent to which leaders foster teamwork, organize and direct project work, manage relationships with external stakeholders (Mohrman *et al*., 1995) and stimulate creativity and innovation. Barry (1991) conducted a detailed qualitative study of engineering and product development teams. He identified four leadership roles that are critical to ensure teams are able to tackle the challenges of R&D work. Yukl (2002) refined this taxonomy to identify four roles: boundary spanning, facilitative, innovation-stimulating leadership as well as directive leadership. Leadership boundary spanning involves the management of external relationships including coordinating tasks, negotiating resources and goals with stakeholders as well as scanning for information and ideas. Facilitative leadership measures whether the leader encourages an atmosphere conducive to teamwork ensuring team interactions are equitable and safe; encouraging participation, sharing of ideas and open discussion of different perspectives. A leader who acts as an innovator envisions project opportunities and new approaches by questioning team assumptions and challenging the status quo. Directive leaders' drive structured and ordered performance of project work by communicating instructions, setting priorities, deadlines and standards. We adopt this framework as it provides a parsimonious and practical description of the specific leadership behaviours required to build task orientation, ensure team cooperation, foster innovation, as well as manage external relations. Whilst studies have independently isolated the importance of the roles (e.g. Waldman and Atwater, 1994; Kim *et al*., 1999; Kim and Yukl, 1995) no study has tested the taxonomy developed by Barry (1991) and Yukl (2002). Thus given that Barry's (1991) research indicates each role serves a critical function maintaining a team's operation we propose the four roles will be collectively related to performance:

H1. Leadership role performance is positively associated with project performance.

2.2. Boundary spanning

R&D teams are dependent on the external environment for resources, information and support (Ancona and Caldwell, 1992). Boundary-spanning communication includes politically orientated communication that increases the resources available to the team, and networking communication, which expands the amount and variety of information available to a team (Brown and Eisenhardt, 1995). Ancona and Caldwell (1992) conducted a longitudinal study of 45 product development teams. Boundary spanning, involving political activities such as negotiating and lobbying resources, was a significant predictor of research managers' ratings of performance. Keller (2001) found that based upon a sample of 93 research teams that external communication was a significant predictors of managers ratings of technical quality ($\beta = 0.56$), budget ($\beta = 0.29$) and schedule performance ($\beta = 0.24$). The researchers concluded that teams and their members must engage in a range of boundary spanning activities in order to succeed. Thus we put forward the following hypothesis:

H2. Team boundary spanning is positively associated with project performance.

2.3. Communication safety

Communication processes form the building block of effective teamwork, providing the means to exchange information, ideas and different perspectives amongst team members (Hoegl and Gemuenden, 2001). The study of team processes reflects a rich tradition of research, spawned by human relations theories that identified social interaction as a potent motivator and stimulus for innovation. Kivimaki *et al.* (2000) studied 493 employees. Of the eight different facets of organizational communication measured, participative communication was the strongest predictor of innovation effectiveness ($r = 0.60$) and patents produced ($r = 0.19$). Participation leads to a more complete understanding of potential problems, as useful information is shared, leading to the cross fertilization of ideas, spawning innovation (Mumford and Gustafson, 1988). Open discussion of

© Blackwell Publishing Ltd 2004

diverse viewpoints creates uncertainty about the adequacy of one's position, curiosity and information seeking to understand the contrary view. Because people understand opposing ideas and information, they are able to see the limitations in their views and incorporate other perspectives, leading to high quality decisions (Tjosvold, 1985). Conversely, power conflicts between individuals and coalitions within the team, may impair team work as anxiety impairs cognitive processing of complex information (Amason and Sapienza, 1997), as individuals become less receptive to the ideas of others and energy devoted to task work is directed towards resolving conflict (West *et al.*, 1998). Donnellon (1986) conducted a series of detailed in-depth studies of product development teams. She observed cliques and coalitions between different functional groups were responsible for costly delays, and integration difficulties. Based upon previous research we propose the following:

H3. Communication safety is positively associated with project performance.

2.4. Reflection

Research studying knowledge work teams (cf. Edmondson *et al.*, 2001; West, 2000) has identified the importance of reflection in complex problem solving activities. West (2000) calls this construct team reflexivity and describes it as the extent to which team members collectively reflect upon the team's tasks (including objectives, strategies) or processes and adapt them to current or anticipated circumstances. Dunbar (1996) studied four renowned science laboratories tracing the processes underlying scientific discoveries. He found that scientific discoveries most commonly occurred when groups reflected on potential causes for negative or inconsistent findings. The findings mirror West's construction of reflexivity (cf. West *et al.*, 1997) whereby reflection was more effective if it occurred in teams, as individuals tended to discount anomalous findings. Secondly, reflection stimulated the reframing of individuals cognitive representations of tasks and questioning of assumptions, leading to the proposal of alternative, novel and innovative approaches. Thus we propose the following:

H4. Team reflexivity will be positively associated with project performance.

2.5. Task communication

The inclusion of the task communication factor reflects a rigorous body of research developed by the goal setting literature. We examine four aspects of task communication: clarity of objectives, feedback, information transmission and understanding of customer requirements. Clarity of objectives and the quality of feedback have been consistently identified as a significant predictor of performance. Thamhain and Wilemon (1987) examined 60 potential driving forces and barriers to the performance of engineering teams. Proper direction and leadership ($r = 0.55$) was one of the strongest drivers of performance, whereas unclear objectives ($r = -0.40$) were one of the strongest inhibitors of performance. A subsequent study conducted by Thamhain (1996) based upon 74 self-directed engineering teams replicated this finding. Communication of project objectives was one of the strongest correlates of performance. Clear direction setting enables the focused development of ideas which can be assessed with greater precision than if unclear (West and Anderson, 1996). Effective information transmission is important to provide teams with sufficient knowledge to enable informed selection of project strategies (Hackman, 1990). R&D teams that commonly perform non-routine and non-repetitive tasks require frequent task communication to provide a comprehensive and complete understanding of complex inter-related activities. In particular, interaction with project customers ensures a clear understanding of what is needed to meet the customer's requirements, how it is to be achieved and how successfully it has been achieved. Zirger and Maidique (1990) compared 70 successful products and failures. They found that customer-team interaction differentiated project success enabling focused and directed work, providing opportunities for learning. Thus consistent with research and theory we put forward the following hypothesis:

H5. Task communication is positively associated with project performance.

2.6. Functional leadership

The functional leadership model proposes that effective leadership stimulates teamwork, which in turn enhances project performance. Applying this framework to our research, we predict that the relationship between leadership role performance and project performance will be mediated by similar communication behaviours. Thus a facilitative leader will encourage communication safety, which in turn will enhance performance. Likewise a directive leader will enhance project

 © Blackwell Publishing Ltd 2004

performance largely through their impact on task communication. Consistent with this rationale we put forward the following hypothesis:

H6. Team communication mediates the relationship between leadership role performance and project performance.

3. Method

Four organizations participated, providing 56 R&D teams. Two of the companies were R&D divisions of large publicly-listed mining and chemical manufacturing organizations, whilst the other two were government R&D organizations. The analyses report a one-year longitudinal study, with repeated mail outs every four months. Analyses were conducted on team level data collected over four time periods. Teams (both leaders and team members) rated leadership role performance and team communication. Teams, research managers and customers rated project performance. At the beginning of the study there were 56 teams, comprising 294 team members and 56 leaders (providing 78% response rate). Fifty-three teams participated at mail-out two (four months later), 41 teams participated at mail-out three (eight months later), and 37 teams participated at mail-out four (one year later). Twenty customers rated performance at mail-out two. However response rates were too low for statistical analyses at mail-out four. Thirty-one research managers rated project performance at mail-out one, whilst 33, 24 and 22 research managers rated performance four months later (mail-out two), eight months (mail-out three) and one year later (mail-out four).

On average, managers supervised six project leaders (s.d. 4.60). The mean team size was eight people (s.d. 4.72). Based on the classification system reported by Katz and Tushman (1979) most teams in the study were engaged in applied research (39%) or product development projects (33%). Basic and technical services projects each accounted for 4% of the sample. A fifth (20%) of the projects spanned a range of R&D types.

3.1. Questionnaire measures

Leadership role performance was assessed on a seven-point scale, '1' corresponded to 'not at all well', '4' 'moderately well,' and '7' 'extremely well'. Team communication items were rated on a 5-point Likert-type scale ranging from strongly disagree '1' to strongly agree '5'. Means, standard deviations, chronbach alphas (α), and inter-rater agreement (rwg) are reported in Table 1 and all items are listed in Appendix A. Confirmatory factor analysis was used to assess construct measurement. A two-stage analysis was conducted based upon individual level data ($n = 350$) from team members and project leaders, consistent with the procedures outlined by Bollen (1989). The first stage examined the measurement properties of scales within each factor. The second set of analyses assessed the latent variable equation of the original model, i.e. treating the team factors as though they were observed variables, examining the inter-relationship between scales comprising the five factors and the correlation between factors. Indices of model fit were acceptable: Joreskog AGFI = .86, Bentler CFI = .93. The team boundary spanning scale, based upon the work of Ancona and Caldwell (1992) demonstrated sound measurement properties: AGFI = 0.92, CFI = 0.93. The communication safety factor demonstrated sound measurement properties: AGFI = 0.93, CFI = 0.96. The reflexivity measure displayed acceptable fit indices: AGFI = 0.85, CFI = 0.92. Task communication demonstrated adequate fit; AGFI = 0.88, CFI = 0.95. Confirmatory factor analysis indicated that the second order factor structure was an acceptable fit to the data (AGFI = 0.83, CFI = 0.93). Comparison of three and four factor models revealed they were not a better fit to the data. The largest correlation was found between facilitative leadership and communication safety ($r = 0.67$). Whilst the correlations between some scales were large, leading to concerns regarding discriminant validity of the scales, the inter-correlated nature of team communication is consistent with the presence of conceptual relationships between different aspects of team communication. In summary, the first analyses confirmed scale measurement was acceptable and the second step showed that the latent variable model loadings were also acceptable, providing sufficient evidence to identify the model as a whole, consistent with the criteria outlined by Bollen (1989).

The team performance measure included seven items covering whether the team had chosen the 'best available strategies for meeting project goals' and 'succeeded in meeting project objectives/milestones'. Correlations between different stakeholders' assessments of project performance were averaged across time to increase the sample size for these analyses. Team ratings were significantly correlated with customer ratings ($r = 0.44$)

© Blackwell Publishing Ltd 2004

Table 1. Means, standard deviations, reliabilities, rwgs of leadership and team communication measures and correlations between stakeholders' ratings of performance.

	M (S.D.)	α	rwg	1	2	3	4	5	6	7	8	9	R^2	N
1. Leadership role performance†	4.98 (0.58)	0.92	0.81											
2. Boundary spanner	5.10 (0.60)	0.84	0.74	**0.73****										
3. Facilitator	4.96 (0.85)	0.79	0.70	**0.81****	**0.39****									
4. Innovator	4.78 (0.80)	0.65	0.67	**0.80****	**0.45****	**0.51****								
5. Director	5.02 (0.77)	0.77	0.73	**0.85****	**0.63****	**0.62****	**0.62****							
6. Team boundary spanning	3.94 (0.30)	0.60	0.79	**0.31***	**0.31****	**0.22***	**0.30***	**0.22***						
7. Communication safety	3.91 (0.36)	0.88	0.84	**0.49****	0.18	**0.67****	**0.34****	**0.31****	0.30					
8. Team reflexivity	3.10 (0.32)	0.76	0.82	**0.35****	0.15	**0.42****	**0.23***	0.27	**0.32****	**0.43****				
9. Task communication	3.55 (0.41)	0.88	0.75	**0.53****	0.21	**0.56****	**0.42****	**0.49****	**0.25***	**0.31****	**0.59****			
Performance: Team, 4 months	3.84 (0.37)	0.89	0.93	**0.36****	**0.30***	**0.49****	**0.28***	0.21	**0.31****	**0.43****	**0.34****	**0.40****	0.67	53
Performance: Team, 1 year	3.90 (0.45)	0.91	0.93	0.42**	**0.49****	**0.54****	**0.41***	**0.29***	0.07	**0.46****	0.45**	**0.44****	0.65	37
Performance: Customer, 4 months	3.95 (0.66)	0.91	n.a	0.34	0.34	0.33	0.34	0.19	−0.12	**0.54***	0.23	0.12	0.50	20
Performance: Research manager, 4 months	3.87 (0.60)	0.87	n.a	**0.37***	0.27	0.21	**0.38***	**0.33***	0.23	0.14	0.05	0.22	0.48	33
Performance: Research manager, 1 year	4.16 (0.58)	0.83	n.a	0.30	0.35	0.25	**0.36***	0.03	−0.04	0.05	−0.27	0.05	#	22

†Leadership role performance composite of four roles. **significant at 0.01 level; *significant at 0.05 level.
#not computed due to bi-modal distribution. Adjusted R^2 values calculated according to Browne (1975) providing a conservative measure of adjusted R^2 values for samples < 40 (Cattin, 1980).

but not research managers ($r = 0.19$). Research managers ratings were significantly correlated with customer ratings ($r = 0.44$).

4. Results

Analyses were conducted on team level data.[1] We divide the results according to the three sets of performance ratings as support for the individual hypotheses varies across the three stakeholders. Table 1 reports the correlations between leadership roles and team communication measures at mail-out one as well as their relationship with project performance measured four months later (mail-out one) and one year later (mail-out four).

4.1. *Team ratings of performance*

Across all four-time periods, four of the five communication processes: leadership role performance (H1), communication safety (H3), team reflexivity (H4) and task communication (H5), were significantly correlated with team performance. When the five factors were entered simultaneously into a regression model, task communication was a significant determinant of team performance at mail-out two, $F(5, 47) = 2.08$, $p < 0.05$. However the other factors were not significant.

Based on team ratings of performance hypothesis five received the strongest support. Task communication was the strongest predictor of performance. Least support was provided for hypothesis two. While leadership boundary spanning was related to team performance across different time periods, team boundary spanning was only significantly correlated with team performance measured at mail-out two. We used two approaches to compare and contrast leadership and team boundary spanning. First, we examined leaders' perceptions of leadership boundary spanning role performance, and team members (only) ratings of team boundary spanning. Almost no association ($r = 0.03$) was found between leaders ratings and team boundary spanning. To isolate potential antecedents of leadership and team boundary spanning, we examined measures of project complexity (i.e. perceived difficulty, obstacles and type of R&D) as well as leader attributes (i.e. leader tenure both years in the organization and as a project leader, number of projects previously supervised, perceived control over project decisions) and factors (team tenure,

© Blackwell Publishing Ltd 2004

co-location and team size). Amongst these measures older project leaders reported higher levels of leadership boundary spanning ($r = 0.33$, $p = 0.09$, $N = 26$), as did leaders who were responsible for resolving disputes and project decisions ($r = 0.32$, $p < 0.05$). Secondly we compared the frequency of leader and team communication with (i) customers, (ii) managers and (iii) technical experts. This scale was re-coded as an ordinal measure (i.e. a leader who speaks to a customer on a daily basis, interacts 20 times per month). Project leaders interacted more frequently with external stakeholders. For example the majority (62.5%) of project leaders spoke to project customers at least on a weekly basis, whereas less than a quarter (22.4%) of team members spoke to customers on a weekly basis. Sixty percent (62.8%) of project leaders and slightly less than half (48.1%) of team members spoke with colleagues external to the project on a weekly basis. Little or no relationship was found between the frequency of leader and team communication with external stakeholders. The largest correlation observed was a negative association between the frequency of leader and team interaction with technical experts ($r = -0.16$).

4.2. Hypothesis 6: Functional leadership

We conducted three sets of regression analyses to test for the presence of moderated relationships as well as the functional leadership model. The first analyses examined whether an alternate moderated model was a better fit to the data than the functional leadership model. Moderated regression models did not explain more variance in project performance (measured four months later) than simple regression models. Thus the analysis did not provide support for the proposition that leadership and team communication factors interact to predict performance.

The second regression analyses tested the functional model of leadership assessing whether the relationship between leadership role performance and team performance (measured four months later) was mediated by related team communication factors. According to Baron and Kenny (1986) three conditions are required before one can test for mediation: the independent and mediator variables must be correlated (all leader roles and team factors met this condition), the independent and dependent variables must be correlated (all leader roles with the exception of the director role which was near significantly correlated ($r = 0.21$) met this condition) and the mediator and dependent variables must be correlated (all team factors met this criteria). Establishing mediation requires that the effect of an independent variable be less when the mediator is included in the regression equation than when the mediator is not included (Keller, 2001). Table 2 reports the regression results assessing the variance explained by the independent and mediator variables separately and together in the mediated model on team performance measured at mail-out two. The change in the adjusted R-square between the mediator and mediated model (i.e. the independent variable and the mediator) is reported in the last column. Entering leadership and team boundary spanning simultaneously into the model reduced the variance explained by leadership boundary spanning. However when entered simultaneously both measures were not significant indicating that whilst team boundary spanning attenuated the relationship with performance it did not completely

Table 2. Tests of mediation: does team communication mediate the relationship between leadership role performance and team performance.

	Direct		Mediator only		Mediated Model		
	β	Adjusted R^2	β	Adjusted R^2	β	Adjusted R^2	Δ Adjusted R^2
Boundary spanner	0.34*	0.10		0.08	0.26	0.12	0.04
Team boundary spanning			0.31*		0.22		
Boundary spanner	0.34*	0.10		0.17	0.21	0.19	0.02
Task communication			0.43**		0.35*		
Facilitator	0.49**	0.22		0.18	0.36*	0.23	0.05
Communication safety			0.44**		0.19		
Innovator	0.28*	0.06		0.11	0.20	0.13	0.02
Reflexivity			0.36**		0.31*		
Director	0.22	0.03		0.17	−0.09	0.15	−0.02
Task communication			0.42**		0.44**		

Team ratings of team performance measured at mail-out two. *significant at 0.05, **significant at 0.01.

© Blackwell Publishing Ltd 2004

explain all the variance, consistent with partial mediation only. When facilitative leadership and communication safety were simultaneously entered into regression analyses, facilitative leadership remained a significant predictor of performance. Entering innovative leadership and team reflexivity simultaneously reduced the variance explained by innovative leadership behaviour, indicating that reflexivity mediated the relationship with performance. As stated previously the director role was a non-significant predictor of team performance, $F(1,\ 50) = 2.52$, $p = 0.12$. Entering project clarity removed the variance explained by leader role performance, such that the beta coefficient was weak and negative.

We performed cross-lagged analyses to assess the link between leadership and team communication. For example, we tested whether facilitative leadership, controlling for communication safety at mail-out one, predicted communication safety one year later (or vice versa). These analyses tested whether effective leadership behaviours led to positive team communication or vice versa, and are reported in Table 3. No consistent patterns of results were evident for the director role and task communication. Facilitative leadership was a significant predictor of communication safety one year later, even after controlling for communication safety at mail-out one, $F(2,\ 34) = 17.91$, $p < 0.01$. Communication safety did not predict facilitative leadership. Similarly the innovator role was a significant predictor of team reflexivity, controlling for reflexivity at mail-out one, $F(2,\ 34) = 3.81$, $p < 0.05$. Team reflexivity did not predict innovator role performance. Thus the results demonstrate that the facilitator and innovator roles stimulate communication safety or team reflexivity rather than the converse.

4.3. *Customer ratings of performance*

Adjusted *r*-squares indicated that the model explained a quarter of the variance in team performance (28%). When all factors were entered simultaneously into a regression model, only communication safety were a significant predictor of team performance, $F(5,\ 15) = 1.28$, $p < 0.05$. Communication safety ($r = 0.54$) and all three scales comprising this factor: participative decision making ($r = 0.45$), power conflict ($r = 0.47$) and open discussion ($r = 0.39$) were significantly correlated with team performance at mail-out two.

4.4. *Research managers' ratings of performance*

The team communication model explained 27% of the variance in ratings of team performance at mail-out two. Overall leadership role performance($r = 0.37$), the innovator ($r = 0.38$) and the director role ($r = 0.33$), were significantly correlated with team performance at mail-out two. Within the task communication factor, the clarity of objectives scale was significantly correlated with team performance across the first three time periods (mail-out one $r = 0.31$, mail-out two $r = 0.29$, mail-out three $r = 0.35$).

5. Discussion

While the model of team communication explained a substantial proportion of the variance in project performance over time, different communication factors predicted different stakeholders' assessments of performance (see summary Table 4). Like Hooijberg and Choi's (2000) results, the findings strikingly illustrate that there are systematic differences in the importance of different

Table 3. Tests of the link between leadership and team communication: leader roles and team communication factors as determinants of team and leadership communication one year later.

I.V. (M1)	β	Adjusted R^2	D.V. (M4)	β	Adjusted R^2	D.V. (M4)
Boundary spanner	0.34*	0.38	Team boundary spanning	0.81**	0.58	Boundary spanner
Team boundary spanning	0.39*			−0.07		
Facilitator	0.31*	0.50	Communication safety	0.58**	0.48	Facilitator
Communication processes	0.50**			0.19		
Innovator	0.35*	0.14	Reflexivity	0.38*	0.16	Innovator
Reflexivity	0.13			0.16		
Director	0.14	0.44	Task communication	0.60**	0.37	Director
Task communication	0.60**			0.08		

Significant at *0.05 level, **significant at 0.01 level.

© Blackwell Publishing Ltd 2004

Table 4. Support for different hypotheses across stakeholders.

	Stakeholder		
Factor	Team	Customer	Research manager
H1.Leadership role performance	✔	?	✔
H2.Team boundary spanning	?	×	?
H3.Communication safety	✔	✔✔	×
H4.Reflexivity	✔	?	×
H5.Task communication	✔✔	×	✔

✔✔ = strong support, ✔ = supported, ? = weak support, × = not supported.

factors across different raters. Customers' ratings of performance provided the strongest support for hypothesis three. Communication safety was the only significant predictor of customer ratings of project performance. Whereas research managers' ratings of performance provided the strongest support for hypothesis five; task communication was the most consistent predictor of performance. Mixed support was provided for the functional leadership framework. We discuss different stakeholder perspectives, the distinction between leadership and team activities future research directions as well as the managerial implications following.

Whilst there has been an explosion in interest and research examining stakeholder management (Jones and Wicks, 1999) much of this attention has been directed towards the strategic level of the firm (cf. Ogden and Watson, 1999). Team studies that have used multiple performance measurements (cf. Campion *et al*., 1996; Hoegl and Gemeuneden, 2001) have largely ignored the theoretical implications of divergent sets of ratings. As a consequence there is a theoretical gap within the research. Thus we possess limited theoretical basis to explore systematic influences on the orientations of different stakeholders. We acknowledge this as both a limitation of our research and the wider teams' field. Applying stakeholder theories to the results leads us to conclude that key measures of effectiveness from an organizational perspective are those that encourage efficiency and ordered task performance. Thus it was not surprising that task communication was the strongest predictor of team members' and research managers' ratings project performance. Task communication is likely to ensure orderliness and structure, facilitating the efficient use of resources and therefore enhances operational effectiveness. Efficiency, whilst valuable from an organizational perspective, may not necessarily spark innovation and the development of competitive processes and ideas that are crucial from a customer's perspective, and thus is unlikely to be related to customers' ratings of performance.

Whilst a stakeholder perspective provides accurate predictions for the relationship between task communication and organizational performance, the link between communication safety and customer ratings of performance is counterintuitive. Further, the absence of any association between team boundary spanning or customer interaction and customer ratings of performance runs contrary to a stakeholder perspective, which proposes that activities which facilitate the team and customer bond are likely to be those most valued by the customer. The results raises questions as to how and when do we know stakeholder perspectives should be applied? And when are alternate theoretical perspectives more appropriate? Further research is needed to a priori test and contrast stakeholder theories with more established theoretical frameworks, such as the goal setting literature or human relations theories, to understand when stakeholder based theories, provide more accurate predictions of empirical relationships.

We conclude that Zaccaro *et al*. (2001) functional leadership framework is an appropriate model for understanding the relationship between some types of leadership and team communication. For example, leadership and team boundary spanning activities are relatively independent processes, providing limited support for the notion that team members role model leaders by engaging in high levels of boundary spanning. In contrast, consistent with the predictions of the functional leadership model, team reflexivity mediated the link between innovative leadership and team performance, whilst cross-lagged analyses demonstrated innovative leadership behaviour led to team reflexivity. Thus team leadership activities may act in a differential fashion, i.e. some but not all leadership activities may drive team communication. Project leaders do not necessarily serve as role models leading team members to engage in certain types of communication, nor

© Blackwell Publishing Ltd 2004

does task focused leadership necessarily lead to effective task communication and project clarity. Rather, the data suggests that the functional leadership model is the most appropriate framework to study leadership roles that stimulate positive processes, encouraging interaction and reflection.

5.5. Managerial implications

Interview data from a leader responsible for a politically tumultuous project poignantly illustrates why different stakeholders may hold competing perspectives:

> We didn't get stakeholders completely right … The stakeholders included the end-user customer, it included the plant that had to manufacture the product, the distribution and logistics system people … When you pack products into cases the more you put in a case, the harder it is squeeze them in and the more work an operator has to do, so they don't like us trying to get a lot of product into a case. But, the supply and transport people, they love it. At the end of the day, whatever you do, every stakeholder will perceive that their interests are being compromised and they're probably right.

The example of the new packaging product illustrates an inherent structural conflict where different functions hold competing goals. Companies should ensure that different functions performance targets are weighted towards superordinate goals, i.e. the release of a successful new product as opposed to individual cost reduction targets. Further, managers can enhance commitment to common goals by lobbying executives to visibly support the development of a new product, and by explicitly linking new product release with key business and employee objectives, i.e. increased revenue and in turn enhanced resources and job security (Mohrman *et al*., 1995). Similarly, the identification of external threats, such as competitor new product releases, will increase salience of common performance targets. Having worked to align the structural incompatibilities, Thomas (1992) suggests a range of problems solving behaviours to facilitate trust, goodwill and cooperation whilst facilitating discussion of contentious issues. These problem solving behaviours include frequently questioning, summarizing, evaluating and testing understandings, keeping the other party informed of ones actions, whilst avoiding the use of unfavorable statements towards the other party or their interests, as well as adversarial comments which defend or attack an approach and stating disagreements without first providing reasons. Laying the groundwork for integrative negotiation will increase the likelihood that parties accurately exchange information, underlying concerns and openly discuss probable outcomes of alternative courses of action. Finally senior management sign-off of agreements within companies will provide formalized evidence of senior management buy-in.

Managers should encourage leaders to develop innovative role behaviours and boundary spanning leadership. Whilst training programs may teach leaders how to stimulate debate and act as a devils advocate, attention to systemic issues is essential to provide sustained behavioural change. Performance appraisal processes supporting innovation, i.e. requiring leaders to provide evidence of innovations developed, are necessary to provide the impetus for lasting behavioural changes. We suggest organizations can enhance a leader's boundary spanning ability by implementing mentoring programs which socialize leaders to organizational norms, practices and develop leaders' influencing and championing skills. Whilst improvements in the transparency and objectiveness of resource mechanisms may 'on paper' greatly diminish the importance of boundary spanning behaviours, in many organizations important decisions will continue to be made through often informal and hidden networks. Similarly, complex technical information is often best disseminated by face-to-face interaction and in the worst-case scenario, these formalized protocols serve to legitimize hidden decisions and hide tacit knowledge. Therefore we suggest leaders should be encouraged to build their networks. In particular the use of in-house research forums and networking events particularly for peer groups provide an efficient means of developing leaders' intra-company links and the potential pool of knowledge sharing resources.

Communication safety was the strongest most consistent predictor of customer ratings of project performance. Given that project customers are the ultimate arbiters of performance, we propose practitioners should direct much of their attention towards facilitating these processes. In particular, we suggest a step-wise approach to progressively provide the conditions, good will

 © Blackwell Publishing Ltd 2004

and trust necessary for communication safety. Initially, practitioners should identify whether there is conflict and rivalry amongst team members. Should conflict be evident, leaders should stress the importance of common targets, encouraging team members to focus on issues not personalities and if conflict continues consider altering a team's membership. Having addressed issues of power conflict, practitioners should encourage all team members to participate in team discussions. Researchers have found that humor (Avolio *et al.*, 1999) and positive affect (West, 1997) encourages participation as well as enhancing the creativity of individuals suggestions. Having targeted participation, facilitators should increase the openness by which individuals express their opinions, in particular discouraging negative evaluations of ideas. Rather if there are differences of opinion, leaders should encourage team members to suggest alternate positions and illustrate why these may be preferable.

In conclusion, the study re-iterates the importance of communication in driving innovation and project performance. However our findings add that perceptions of performance are relative as different factors predict different stakeholder's ratings of performance. Thus there is an urgent research and practical need to understand why stakeholders adopt different perspectives. Further research should compare and contrast different interventions to test how effective they are in developing and maintaining consensus between different stakeholder groups.

Acknowledgement

This study was supported by a grant from the Australian Research Council to Professor Leon Mann. The authors acknowledge the insightful feedback and advice of Michael West, Felix Brodbeck and Amy Edmondson.

References

Allen, T.J. (1984) *Managing the Flow of Technology: Technology Transfer and the Dissemination of Technological Information within the R&D Organization* (2nd ed.). Cambridge MA: MIT Press.

Amason, A.C. and Sapienza, H.J. (1997) The effects of top management team size and interaction norms on cognitive and affective conflict. *Journal of Management*, **23**, 4, 495–516.

Ancona, D.G. and Caldwell, D.F. (1992) Bridging the boundary: external process and performance in organizational teams. *Administrative Science Quarterly*, **37**, 634–665.

Avolio, B.J., Howell, J.M. and Sosik, J.J. (1999) A funny thing happened on the way to the bottom line: Humor as a moderator of leadership style effects. *Academy of Management Journal*, **42**, 2, 219–227.

Baron, M.R. and Kenny, D.A. (1986) The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, **51**, 6, 1173–1182.

Barry, D. (1991) Managing the bossless team: lessons in distributed leadership. *Organizational Dynamics*, **20**, 1, 31–47.

Bollen, K.A. (1989) *Structural Equations with Latent Variables*. New York: John Wiley.

Boutty, I. (2000) Interpersonal and interaction influences on informal resource exchanges between R&D researchers across organizational boundaries. *Academy of Management Journal*, **43**, 1, 50–65.

Brodbeck, F.C. (2001) Communication and performance in software development projects. *European Journal of Work and Organizational Psychology*, **10**, 1, 73–94.

Brown, S.L. and Eisenhardt, K.M. (1995) Product development: past research, present findings and future directions. *Academy of Management Review*, **20**, 343–378.

Browne, M.W. (1975) Predictive validity of a linear regression equation. *British Journal of Mathematical and Statistical Psychology*, **28**, 79–87.

Campion, A.M., Papper, E.M. and Medsker, J.M. (1996) Relations between work group characteristics and effectiveness: a replication and extension. *Personnel Psychology*, **49**, 429–452.

Cohen, S.G. and Bailey, D.E. (1997) What makes teamwork: group effectiveness research from the shop floor to the executive suite. *Journal of Management*, **23**, 3, 239–290.

Donnellon, A. (1986) Team work: linguistic models of negotiating differences. In: Sheppard, B., Lewicki, R. and Bies, R. (eds), *Research on Negotiations in Organizations* (Vol. 4). Greenwich CT: JAI Press.

Dunbar, K. (1996) How scientists really reason: scientific reasoning in real-world laboratories. In: Sternberg, R.J. and Davidson (eds), *The Nature of Insight*. Cambridge MA: MIT Press, pp. 365–395.

Edmondson, A.C., Bohmer, R.M. and Pisano, G.P. (2001) Disrupted routines: team learning and new technology implementation in hospitals. *Administrative Science Quarterly*, **46**, 685–716.

Fiedler, F.E. (1964) A contingency model of leadership effectiveness. In: Berkowitz, L. (ed.), *Advances in Experimental Social Psychology*, **1**, 149–190. New York: Academic.

Hackman, J.R. (1990) *Groups that Work (and those that Don't)*. San Francisco CA: Jossey-Bass.

© Blackwell Publishing Ltd 2004

Hoegl, M. and Gemeunden, H.G. (2001) Teamwork quality and the success of innovative projects: a theoretical concept and empirical review. *Organization Science*, **12**, 4, 435–449.

Hooijberg, R. and Choi, J. (2000) Which leadership roles matter to whom? An examination of rater effects on perceptions of effectiveness. *Leadership Quarterly*, **11**, 3, 341–364.

Jones, T.M. and Wicks, A.C. (1999) Convergent stakeholder theory. *Academy of Management Review*, **24**, 2, 206–221.

Katz, R. and Tushman, L.T. (1979) Communication patterns, project performance, and task characteristics: an empirical evaluation and integration in an R&D setting. *Organizational Behaviour and Human Performance*, **23**, 139–162.

Keller, R.T. (2001) Cross-functional project groups in research and new product development: diversity, communications, job stress and outcomes. *Academy of Management Journal*, **44**, 3, 547–555.

Kerr, S. and Jermier, J.M. (1978) Substitutes for leadership. *Organizational Behaviour and Human Performance*, **22**, 375–403.

Kim, Y., Min, B. and Cha, J. (1999) The roles of R&D team leaders in Korea: a contingent approach. *R&D Management*, **29**, 2, 153–165.

Kim, H. and Yukl, G. (1995) Relationships of managerial effectiveness and advancement to self-reported leadership-reported leadership behaviours from the multiple-linkage model. *Leadership Quarterly*, **6**, 3, 361–377.

Kivimaki, M., Lansisalmi, H., Elovainio, M., Heikkila, A., Lindstrom, K., Harisalo, R., Sipila, K. and Puolimatka, L. (2000) Communication as a determinant of organizational innovation. *R&D Management*, **30**, 1, 33–42.

Marks, M.A., Mathieu, J.E. and Zaccaro, S.E. (2001) A temporally based framework and taxonomy of team processes. *Academy of Management Review*, **26**, 3, 356–376.

McCall, M.W. Jr. (1988) Leadership and the professional. In Katz, R. (ed.), *Managing Professionals in Innovative Organizations*. Cambridge MA: Ballinger.

Mohrman, S., Cohen, S. and Mohrman, L. (1995) *Designing Team Based Organizations*. London: Jossey-Bass.

Mumford, M.D. and Gustafson, S.B. (1988) Creativity syndrome: Integration, application and innovation. *Psychological Bulletin*, **103**, 1, 27–43.

Ogden, S. and Watson, R. (1999) Corporate performance and stakeholder management: balancing stakeholder and customer interests in the U.K. privatized water industry. *Academy of Management Journal*, **42**, 5, 526–538.

Pelz, D.C. and Andrews, F.M. (1976) *Scientists in Organizations*. Ann Arbor MI: Institute for Social Research.

Pirola-Merlo, A., Hartel, C., Mann, L. and Hirst, G. (2002) How leaders influence the impact of affective events on team climate and performance in R&D teams. *Leadership Quarterly*, **13**, 5, 561–581.

Shankman, N.A. (1999) Reframing the debate between agency and stakeholder theories of the firm. *Journal of Business Ethics*, **19**, 319–334.

Thamhain, H.J. and Wilemon, D.L. (1987) Building high performing engineering teams. *IEEE Transactions on Engineering Management*, **EM-34**, 3, 130–137.

Thamhain, H.J. (1996) Enhancing innovative performance of self-directed teams. *Engineering Management Journal*, **8**, 3, 31–39.

Thomas, K.W. (1992) Conflict and negotiation processes in organizations. In: Dunetter, M.D. and Lough, L.M. (eds.), *Handbook of Work and Organizational Psychology* (2nd ed.). Palo Alto CA: Consulting Psychologists Press, pp. 651–717.

Tjosvold, D. (1985) Implications of controversy research for management. *Journal of Management*, **11**, 21–37.

Visart, C. (1976) Communication and R&D performance. In: Pelz, D.C. and Andrews, F.M. (eds), *Scientists in Organisations*. Ann Arbor MI: Institute for Social Research.

Waldman, D.A. and Atwater, L.E. (1994) The nature of effective leadership and championing processes at different levels in a R&D hierarchy. *Journal of High Technology Management Research*, **5**, 2, 233–245.

West, M.A. (1997) *Developing Creativity in Organizations*. Leicester: British Psychological Society.

West, M.A. (2000) State of the art: creativity and innovation at work. *Psychologist*, **13**, 9, 460–464.

West, M.A. and Anderson, N.R. (1996) Innovation in top management teams. *Journal of Applied Psychology*, **81**, 6, 680–693.

West, M.A., Borrill, C.S. and Unsworth, K.L. (1998) Team effectiveness in organizations. In: Cooper, C.L. and Robertson, I.T. (eds), *International Review of Industrial Organizational Psychology*. Chichester: John Wiley.

West, M.A., Garrod, S. and Carletta, J. (1997) Group decision-making and effectiveness: unexplored boundaries. In: Cooper, C.L. and Jackson, S.E. (eds), *Creating Tomorrow's Organizations: a Handbook for Future Research in Organizational Behaviour*. Chichester: Wiley, pp. 293–317.

Yukl, G. (2002) *Leadership in Organizations* (5th ed.). New Jersey: Prentice Hall.

Zaccaro, S.J., Rittman, A.L. and Marks, M.A. (2001) Team leadership. *Leadership Quarterly*, **12**, 451–483.

Zirger, B.J. and Maidique, A.M. (1990) A model of new product development: an empirical test. *Management Science*, **36**, 7, 867–882.

 © Blackwell Publishing Ltd 2004

Appendix A. Questionnaire measures; communication factor, scale and items

Leadership role performance
Boundary Spanner. Coordinating the team's task with outside stakeholders. Collaborating with others outside of the team. Scanning the environment inside and outside the organization for ideas/expertise. Negotiating resources for the team. Negotiating project goals with the customer.
Facilitator. Ensuring all team members have the opportunity to express their ideas and opinions. Acting to ensure that conflicts aren't adversely affecting the team or its members. Engaging in activities to build relationships within the team.
Innovator. Challenging team assumptions about problems to ensure the best solution. Initiating new strategies and approaches to team tasks.
Director. Negotiating allocation of tasks with team members. Establishing standards and priorities. Ensuring milestones and deadlines are met.

Boundary spanning
Team Boundary Spanning. When necessary team members consult with individuals who possess knowledge relevant to the project. Team members scan the environment inside and outside the organization for technical ideas and expertise. Team members seek relevant information from across the organization. Team members have access to individuals who possess technical expertise relevant to the project.

Communication safety
Participative Decision Making. If a problem is encountered, members constructively discuss appropriate actions. Important decisions about the project are made by limited numbers of the team. The input of all members is considered. If a problem is encountered all members are involved in its resolution.
Open Discussion. Team members can express their views openly. Ideas are openly evaluated by team members. Team members do not express their opinions for fear of recrimination or conflict. Within the team, members raise questions when confused or uncertain of an issue.
Power Conflict. There is destructive rivalry between members of the team. Members jockey to control the decisions of the team. To obtain resources or support, members form coalitions with others. Meetings are characterized by arguments between different coalitions within the team.

Reflexivity
Task Reflexivity. The team steps back from daily routines to consider whether the methods used are the best available. The team often reviews its objectives. The methods used by the team are often discussed. The team regularly considers whether work performed meets project objectives.
Process Reflexivity. We regularly discuss whether the team is working effectively together. How well we communicate information is often discussed. The way decisions are made in this team is rarely altered.

Task communication
Clarity of Objectives. Team members have a clear understanding of project objectives. Project objectives are understood by all members. There is a lack of clarity concerning project priorities. Project objectives are clearly communicated to all members.
Feedback. The team receives clear feedback regarding the project's performance. Team members receive clear feedback regarding the quality of project work.
Information Flow. Information circulates throughout the team. It is often difficult to get answers to important questions about my work. Team members have access to all the information required to do their work effectively.
Clarity of Customer Requirements. Team members have a clear understanding of the expectations of customer/funding agencies. The team discusses project objectives with customer/funding agencies. Customer/funding agencies provide clear directions concerning desired project outcomes. The team receives frequent feedback from customer/funding agencies.

© Blackwell Publishing Ltd 2004

Note

1. *Moderating effect of R&D type*. We controlled for R&D type as previous research has found that the nature of the research task moderates the relationship between communication and project performance (Brown and Eisenhardt, 1995). For example, Katz and Tushman (1979) found that external communication across organizational boundaries displayed a significantly stronger positive relationship with performance for research compared to development and technical service team. Thus teams were classified according whether they involved research tasks (i.e. basic or applied research) or product development (i.e. product development or technical services). In order to increase stability and sample size, stakeholders' assessments of performance were averaged across the four mail-outs. Consistent with Katz and Tushman (1979) we found that team boundary spanning displayed a positive relationship with team ratings of performance for research teams (team performance $r = 0.71$, $p < .05$) but not product development teams (team performance $r = -0.17$). No other differences were evident according to R&D type across communication factors or different raters.

 © Blackwell Publishing Ltd 2004